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Supply Chain Hubs in Global Humanitarian Logistics

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Introduction[1]

George quietly sipped his coffee and looked around the table at his team of supply chain experts discussing how to configure their four-wheel drive (4WD) vehicle supply chain. In two weeks, he needed to present a supply chain plan to the leadership of the International Humanitarian Organization (IHO). The new ideas being discussed differed from the strategies of other humanitarian logistics groups and he wondered how they would be received.

George Zuleta was the director of the Fleet Management Group (FMG) which was responsible for the IHO's vehicle fleets, most of which were managed as part of the Vehicle Rental Program (VRP) from an office in the United Arab Emirates (UAE). The team was studying trends, recent deployment data, and various optimization model simulations. Their vehicle supply chain plan needed to provide a fast response to disasters and support cost-conscious development programs. In addition, it needed the flexibility to quickly provide hundreds of vehicles to support mega disaster response and recovery while maintaining a low fixed-cost structure to minimize costs in years without disasters and without much funding.

Known for being one of the first IHOs on the scene of a disaster, IHO leadership wanted to continue providing service to match this reputation. In an effort to improve disaster responsiveness for critical items such as food, water and medicine, other IHO logistics teams were moving towards more decentralized supply chain configurations with several regional hubs around the globe. By having more stock pre-positioned in regional hubs around the globe, they hoped to reduce disaster response time, as well as the need to airlift critical items into the disaster area, which was expensive. By utilizing many regional hubs, the transport costs of critical items should be cheaper because these items could simply be shipped from the nearest hub to the disaster area.

While George understood the logic, he wondered how the demand for 4WD vehicles might differ from demand for food and water. Would these differences impact the disaster response time and cost dynamics?

Currently, the FMG supply chain for vehicles was centralized on a global fleet hub located in UAE, where vehicles were received from the manufacturer, outfitted for specific tasks or countries, repaired and maintained between rental assignments, and prepped for final end-of-use sale. The hub in UAE also supported training, full vehicle management, and maintained a reserve pool of vehicles for unexpected disasters.

The FMG had recently started testing out regional hubs in Panama and Indonesia in the hope that this would improve hub proximity to global disasters, rather than relying on the global fleet hub in UAE alone. George knew that other logistics teams would be proposing supply chain configurations with five or more hubs for prepositioning supplies around the globe. While this could reduce disaster response time, George worried that the structural costs to maintain and operate five vehicle hubs would take up most of the FMG budget, particularly in lean years. These hub structural costs were the fixed costs required to maintain a functioning vehicle hub regardless of the number of vehicles being shipped through the hub, e.g., leasing

1 The names of the organization and people are fabricated for this case but are representative of experiences that have occurred within IHOs around the globe.

warehouse and/or secure parking space, purchasing vehicle maintenance lifts, and paying salaries and benefits for the fleet manager and hub staff.

George recalled how in 2008 and 2009, when there were few disasters to support, the IHO had seen a drop in donations, turning lean disaster years into lean donation years (see Exhibit 2). His budget had been cut to the bone. "I always love years without disasters because that is much better for humanity and society overall," he told the team, "But as a humanitarian organization, we really have to watch our costs and make sure we can survive the lean years and be prepared and ready for the next mega disaster in future years."

He described what happened in Haiti after the 2010 earthquake. In the first few weeks, the FMG shipped 10 vehicles to Haiti from a small reserve pool in Panama, which was being tested as a regional vehicle hub location. Since they were pre-positioned so close to Haiti, the FMG provided an immediate disaster response. Additional vehicles were needed, so George had 40 vehicles sent from the manufacturer in Japan to the regional hub in Panama to be outfitted for disaster recovery work. At the same time, he also shipped about half of the vehicles in the reserve pool from the global fleet hub in UAE to Haiti. Both shipments arrived 3-4 months after the earthquake, just as military operations and disaster relief focused organizations were pulling out. In fact, more vehicles were needed 6-18 months after the disaster than immediately after it occurred because of the high demand for vehicles to transport building supplies, construction workers and other volunteers helping rebuild Haiti in the recovery period after the initial disaster response.

Over the next two years, as George shipped many more vehicles to Haiti to support the disaster recovery effort, Haiti became the IHO's ad hoc regional vehicle hub in the Caribbean instead of Panama. Vehicles were sent directly to Haiti from the manufacturer, outfitted for specific tasks, and rented out to various IHO programs for their own recovery initiatives. Leasing warehouse space, procuring vehicle lifts, and managing vehicles from within Haiti was effective in containing costs while providing the services needed. Vehicles purchased from the manufacturer were put into service sooner and the regional hub in Haiti performed vehicle maintenance and repairs. George knew this would have been extremely difficult to perform in local dealerships in Haiti, yet shipping vehicles back to the global fleet hub in UAE for repair, only to be returned to Haiti to finish their rental assignment, would have been extremely expensive. At its peak as an ad hoc regional vehicle hub, Haiti had over 240 VRP vehicles, second only to the global fleet hub in UAE. Looking back, George knew having a vehicle hub in Haiti (the site of the FMG's biggest demand after the mega disaster) had worked well. If anything, he wished he had made Haiti a fully operational vehicle hub sooner.

But would it work in other locations? Should he make optional temporary hubs part of his overall vehicle supply chain configuration plan? George challenged his team to consider this option as they discussed each proposed supply chain configuration, ranging from a single hub in UAE to half a dozen vehicle hubs all over the globe.

Background on the International Humanitarian Organization

With a mission of responding to natural disasters around the globe, the IHO included 190 National Societies in almost every country by 2016. These National Societies were run from each specific country, not under the control of IHO leadership in Switzerland. However, through these National Societies, the IHO could reach 97 million people via long-term services and development programs, and 85 million people via disaster response and early recovery programs each year.

With its headquarters in Switzerland, the IHO had five zone offices and a number of delegations around the world. The international headquarters supported National Societies by coordinating international support before, during, and after large-scale disasters, strengthening their capacity and leadership, raising international donations, and persuading decision makers to act at all times in the interests of vulnerable people.

A board consisting of a president, four vice-presidents, the chairman of the finance commission and 20 National Society representatives governed the IHO. The highest body, the General Assembly, convened every two years, with delegates from all National Societies. The IHO was funded by statutory contributions from National Societies, the delivery of field services to program partners, and voluntary contributions from donors such as governments, corporations, and individuals. Additional funding, especially for disasters, was raised by emergency appeals (via TV, print, and email). The IHO had more than 165,000 branches or local units, over 427,000 paid staff worldwide and 17.1 million volunteers.

Since 2011, the government of the United Arab Emirates had made a significant contribution to the IHO by hosting its logistics unit in Dubai as a part of the International Humanitarian City (IHC). Established as an independent, free zone authority, the IHC is the world's largest and busiest logistics hub for humanitarian aid, with over 55 international agencies and NGOs. This contribution to the IHO was worth an estimated 505,921 Swiss Francs annually (about $551,773) and covered the provision of office space, storage facilities, and utilities. The IHC derived operational efficiency from the customs clearance and other formalities negotiated with UAE. The FMG was one of the IHO's groups to be based out of the IHC.

Background: Fleet Management Group

Vehicles are a crucial link in delivering aid around the globe and to the smooth operation of any IHO activity, ensuring the safety of volunteers and staff and providing a visible reminder of its presence in humanitarian actions around the globe. Vehicles are the second highest overhead expense for the IHO after staff salaries and benefits. Prior to 2000, National Societies struggled to procure and maintain vehicle fleets themselves, within the decentralized vehicle fleet system, each of them determining the size of their fleet and procuring vehicles on their own.

In an effort to optimize fleet management, the IHO established the Fleet Management Group in 2002, headquartered in Dubai. It used a hybrid management model to run its international vehicle rental program (VRP), renting 600-800 vehicles to National Societies and IHO programs around the world. The FMG managed vehicles for over 80 National Societies. Procurement of 4WD vehicles was divided 80%:20% between the FMG global sourcing and the national offices respectively, from an approved FMG budget. This gave national offices greater decision-making power than more centralized models; they could determine their fleet composition as long as it complied with FMG standards. Vehicles purchased nationally were later incorporated into the VRP.

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The FMG procured vehicles directly from Toyota and Nissan (the double sourcing was to avoid a monopoly). Once delivered, they were usually stored at the global fleet hub in UAE or placed in one of the regional hubs, where they were equipped before being sent to the National Societies, which leased them from the FMG at a monthly rate. Allocation and routing of the vehicles was decided at the national and field level and had to be approved by a dispatcher if the country had more than four vehicles. The FMG was in charge of transporting vehicles to the National Societies. It was estimated that a regional vehicle hub cost at least $10,000 per month to cover warehouse and office space, maintaining vehicle lifts, insurance and the salaries of employees. These were "structural" or "fixed" costs because the overheads did not vary with the number of vehicles shipped through the hub.

Programs followed strict maintenance schedules every 5000, 10,000, 20,000, and 60,000kms. Maintenance was carried out by an authorized dealer or a local workshop approved by the National Society, with a mandatory replacement policy of five years or 150,000kms. The 4WD vehicles were disposed of via sale, donation, or scrap. Sales were organized regionally or from the global fleet hub through auction or tender, depending on the market price and exchange rates; 90% of the revenue from each sale was fed back into the FMG budget for the purchase of new vehicles, making the VRP financially self-sustaining. The remaining 10% went to the National Society budget.

Policies regarding fleet management were drawn up by the FMG and summarized in the fleet manual given to all National Societies. Data on each vehicle was sent to the FMG using the centralized FleetWave information software,[2] which was standard software in many IHOs. Another source of data, the driver's logbook, recorded fuel consumption, accidents, maintenance, insurance, kilometres driven and cost per kilometre –key performance indicators of the vehicle. National Societies did not all use the same information system.

Discussion of Supply Chain Configurations

George's team came from different industry backgrounds, which influenced their recommendations and thoughts about the future supply chain configuration.

Sue was in charge of finances, accounting, and procurement of all vehicles for the VRP. She understood that the FMG needed to maintain a low cost structure so that donated money went to beneficiaries rather than overhead expenses. She wanted to keep costs down so that additional vehicles could be purchased and the lease rate to National Societies could be as low as possible.

Malcolm had spent many years on the ground in disaster areas, coordinating vehicle fleets and ensuring they were used for the most urgent needs to benefit the maximum number of people. He was less concerned about cost, as long as vehicles were available and beneficiaries got the help they needed. His main concerns were the ease of use, safety and durability of the vehicles.

2 FleetWave software is sold by Chevin Fleet Solutions, http://www.chevinfleet.com/us/fleet_management_software.asp

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Alex, the newest member of the team, had joined a year ago after her MBA. She had experience in humanitarian operations as a volunteer and part-time staff, but this was her first full-time position with a humanitarian organization. Alex loved to create optimization models and run various numbers and scenarios through these to determine how decisions might impact the FMG's capabilities and financial position in the future.

Since each had different responsibilities within the FMG, they had their own recommendations for how to restructure the FMG moving forward.

Sue, who was responsible for finances and procurement, wanted a low-cost organization: "I hate requesting extra money in years when only a few disasters occur to pay for multiple hubs around the globe. It feels awkward to tell the executive committee that we need $120,000 annually to pay for a vehicle logistics hub, when in lean disaster years we don't really need that hub." She was quick to point out the benefits of using a centralized supply chain with few vehicle hubs: "If we used one central hub we could reduce our hub structural costs, achieve pooling benefits by keeping all emergency response vehicles in a single pool ready to respond to disasters when required, all while keeping tabs on vehicle maintenance, availability, and outfitting much more easily. We also support a lot of long-term development programs for which we plan out vehicle shipments months in advance, so those vehicles can cost effectively be shipped from a centralized vehicle hub right here in UAE." George appreciated this point, because it concerned one of the key differences between commercial and humanitarian supply chains. In 2013, 48% of IHO expenditures went to disaster response, with the remainder going to development programs and supplementary services.

Humanitarian Vehicle Transportation Costs

There were different types of transportation for the FMG vehicles. **Standard transportation** is used to ship vehicles, usually by ocean freight, for development programs and for the recovery and rehabilitation phases in the months following a disaster. The costs are similar to shipping container ocean rates and are proportional to the distance travelled plus a few fees. For example, shipping a vehicle from UAE to Panama costs ~$4800, but from UAE to South Africa would be ~$2000. **Expedited transportation** is used for high-profile disasters when vehicles are needed very quickly and cannot be procured locally or borrowed from a neighbouring National Society. Expedited vehicles are usually flown into the disaster area on chartered cargo jets, allowing vehicles to be dispatched anywhere in the world rapidly, but at a high cost. It costs between $8,000-$30,000 per vehicle to fly vehicles into disaster areas, so the number of expedited vehicles is usually small – typically 5-10 vehicles per disaster once every couple of years. While expensive, this allows the FMG to quickly arrive on site in a disaster area and bridge the gap until more vehicles can be more economically shipped.

Malcolm focused on disaster responsiveness and ensuring there were always vehicles available to quickly respond. He recommended having several vehicle hubs positioned around the globe to quickly service all minor and mega disasters: "If you position the hubs appropriately, you could facilitate a fast disaster response from a relatively nearby vehicle hub. In fact, think how much cheaper transportation costs would be if you only have to ship

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the vehicles down the coast or better yet, just drive the vehicles to a neighbouring country in Africa to reach the disaster location." He pointed out that having more regional hubs around the globe would probably reduce the amount of expensive vehicle expediting transportation required.

Humanitarian Vehicle Demand

Most commercial supply chains are aligned with the strategic factors of the particular firm. If the firm focuses on a fast customer response and short shipping times, they have many warehouse hubs around the globe to service customers, who pay a premium for this fast delivery. Other firms have longer delivery times, but customers know they are getting the lowest possible price. These have fewer supply hubs.

For humanitarian logistics, organizations align their supply chains to deal with development programs, minor disasters, and mega disasters. Development programs are long-term initiatives to reduce suffering and poverty, usually within developing countries, e.g. providing clean water, vaccinations, mosquito nets, farming programs, disease prevention and economic well-being. They are planned well in advance and cost is a primary concern, since money spent on logistics, transportation, and overhead cannot be spent on providing the service.

Disaster response support requires vehicles to be procured in a short time frame and the speed of response is the primary concern. While costs are always monitored, the prime focus is to help beneficiaries after a disaster occurs. Minor disasters require small shipments of vehicles, usually for a few months. Mega disasters are more sporadic but significantly impact the FMG's operations when they occur. A mega disaster could require a massive vehicle ramp-up immediately after the disaster and extend over 20-30 months. These consume up to 25% of the FMG's vehicle fleet and influence every decision the FMG makes. (An example of the vehicle ramp-up after two mega disasters, the 2010 Haiti earthquake and the 2004 Asian Tsunami, can be seen in Exhibit 4).

Alex understood the trade-off between responsiveness and cost and how that related to the current supply chain debate. As the newest member of the team, she felt less strongly about either option as she prepared the optimization models and ran the analysis that George had requested. In addition to running scenarios with 1, 2, 3, 4, and 5 fixed hubs, Alex worked with George to formalize the temporary hub concept so it could also be modelled. This called for a centralized configuration with fewer fixed hubs, but gave the option to open a temporary hub in any mega disaster location, which could operate as a regional hub as long as sufficient vehicles were in that location. A temporary hub would open only after a mega disaster occurred, so any uncertainty surrounding mega disaster location was eliminated. A separate temporary hub could be opened in each mega disaster location if multiple mega disasters were to occur. Thus, temporary hub locations followed the largest source of vehicle demand through the years. If no mega disasters occurred for several years, then the global fleet hub in UAE might be the only vehicle hub in the configuration. This meant fewer vehicle hubs in years without mega disasters to reduce structural costs and increase the benefits of pooling

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reserve vehicles, while the option of temporary hubs provided the flexibility to support a mega disaster response.

Opening a temporary hub was optional. For example, if a mega disaster destroyed all the local ports or occurred in a landlocked country, it would not make a good hub location; a temporary hub could be opened in a neighbouring country with better infrastructure near a large body of water for inbound vehicle shipments. If a mega disaster occurred near a fixed hub (or a temporary hub from a mega disaster the previous year) it could be supplied from the established hub (a new temporary hub would not need to be opened). George and Alex also had several concerns with the temporary hub concept:

1. What if the mega disaster country had poor import/export regulations and there was a lot of 'red tape' to navigate when moving vehicles in and out? This would make for a very bad hub location, since an IHO probably couldn't negotiate prior agreements with every country and every new government that came to power.
2. What about opening and closing costs? While these were unlikely to be significant (office space, a vehicle warehouse or fenced parking lot, and some vehicle lifts for repairs), he couldn't be sure about every situation or every location.
3. Would this hurt disaster responsiveness and require more expedited transportation of vehicles in the first days or weeks after the disaster? It would take 4-5 weeks to establish the temporary vehicle hub, during which vehicles would need to be flown into the disaster location if local vehicles could not be purchased or borrowed.

Lastly, George was concerned with the funds available for projects. Although some donations to the IHO could be spent wherever the need was greatest, most were earmarked for a specific purpose, program, or country. This worried him because in 2010 he had wanted to spend more money sending vehicles to Pakistan to help with flood relief efforts – monsoon rains in July had left a fifth of the country under water and impacted over 20 million people. But as most of the donations in 2010 were earmarked for Haiti disaster relief, he could not spend the money on flood recovery efforts in Pakistan, even though 17 million more people were impacted in Pakistan than in Haiti.

So while the FMG's 2010 budget was one of the largest ever, it was difficult to do anything with it except provide disaster recovery support for Haiti. Finding ways to utilize earmarked funding for the disaster or program as planned, while positioning the FMG for the other needs and future obligations, was thus a prime goal for any new vehicle supply chain configuration.

Deadline Approaching

The FMG had to simultaneously support development programs and provide support for disasters. The vehicle supply chain would need to respond to mega disasters that historically consumed many resources and a large portion of the VRP fleet *and* be able to survive lean years without disasters or much funding. How could George better utilize the earmarked funding he received? Could they pull off this new temporary hub concept? Did they even want to attempt it? Given current vehicle demand and expectations for the future, how centralized or decentralized should the FMG be?

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As George was pondering these questions, Alex came into his office with the simulation analysis results. "I ran all the simulations with 1, 2, 3, 4, and 5 fixed hubs only and with the option for temporary hubs," she said. (Exhibits 5-8) "As a baseline, the simulations use a ten year time horizon, two randomly located mega disasters over that time frame, and random magnitude and location for development program demand. I also simulated having 2-16 mega disasters over a ten year horizon and varying the magnitude of monthly development program demand. I know the FMG only ships 20 vehicles per month on average, but things change as that demand goes up."

George put down his coffee mug and started looking over the simulation results with Alex and the rest of the team. They only had a few short weeks to make their final recommendations.

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Appendix A

Comment on Humanitarian Aid and Humanitarian Logistics

Humanitarian aid is the organized act of supplying materials, knowledge, and assistance to people in need and is usually focused on people whose current government and institutions cannot provide these services. While humanitarian aid is often thought of as food, medicine, water, and other supplies needed to survive in the short term, it also encompasses items required to deliver these supplies. This can include power generators, transport containers, or the items we study in this case: 4WD vehicles. Oftentimes demand and supply strategies for 4WD vehicles can be different from the demand and supply strategies for medicine or food.

When a disaster occurs, particularly in the case of mega disasters, the infrastructure for delivery of aid is impacted. Roads or bridges can be washed away by flooding or hurricane waves, so a delivery trip that would have taken 30 minutes before the disaster might now require hours. A 4WD vehicle may need to drive miles out of the way to get through a still existing pass, use an intact road, or ford a river in a shallow location, so that the required supplies are distributed to beneficiaries who cannot always travel to retrieve supplies. This final act of delivering the supplies into the hands of the beneficiaries is termed "last mile distribution". Last mile distribution can be very difficult immediately after disasters and for months thereafter until bridges are rebuilt, roads are repaired, and storage locations are secured. Donating money, purchasing supplies, and transporting the supplies to the disaster region are only the first part of the challenge. In addition, vehicles are required to transport workers and supplies to rebuild the bridges, roads, and infrastructure system even if no additional food, water, and medicine are required by beneficiaries. It is the last mile distribution challenges that make support items such as 4WD vehicles critical to providing effective humanitarian relief.

When deploying development programs, the IHO usually benefits from better roads and distribution systems than when working in disaster areas. However, without the urgency of disasters, they can easily be neglected unless distribution tools and workers are specifically designated. For example, the donation of 100,000 mosquito nets would benefit many developing countries, but purchasing the nets and shipping them to a specific country is only the first part of the challenge. If they get stuck in a shipping container in a port city, they are of no help. Vehicles and workers are needed to perform last mile distribution to deliver the nets in the country, and there may also be a need for training and education on how to set up and obtain the maximum benefit from the mosquito nets.

Every IHO organization has a different mix of the amount of disaster response and development programs in terms of expenditure and aid delivered. The International Federation of the Red Cross (IFRC) is divided 50-50 between disaster and development. The International Committee of the Red Cross (ICRC) is more focused on disaster response, and the World Health Organization (WHO) is more focused on development programs.

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Exhibit 1

Number of People Impacted by Disasters by Country

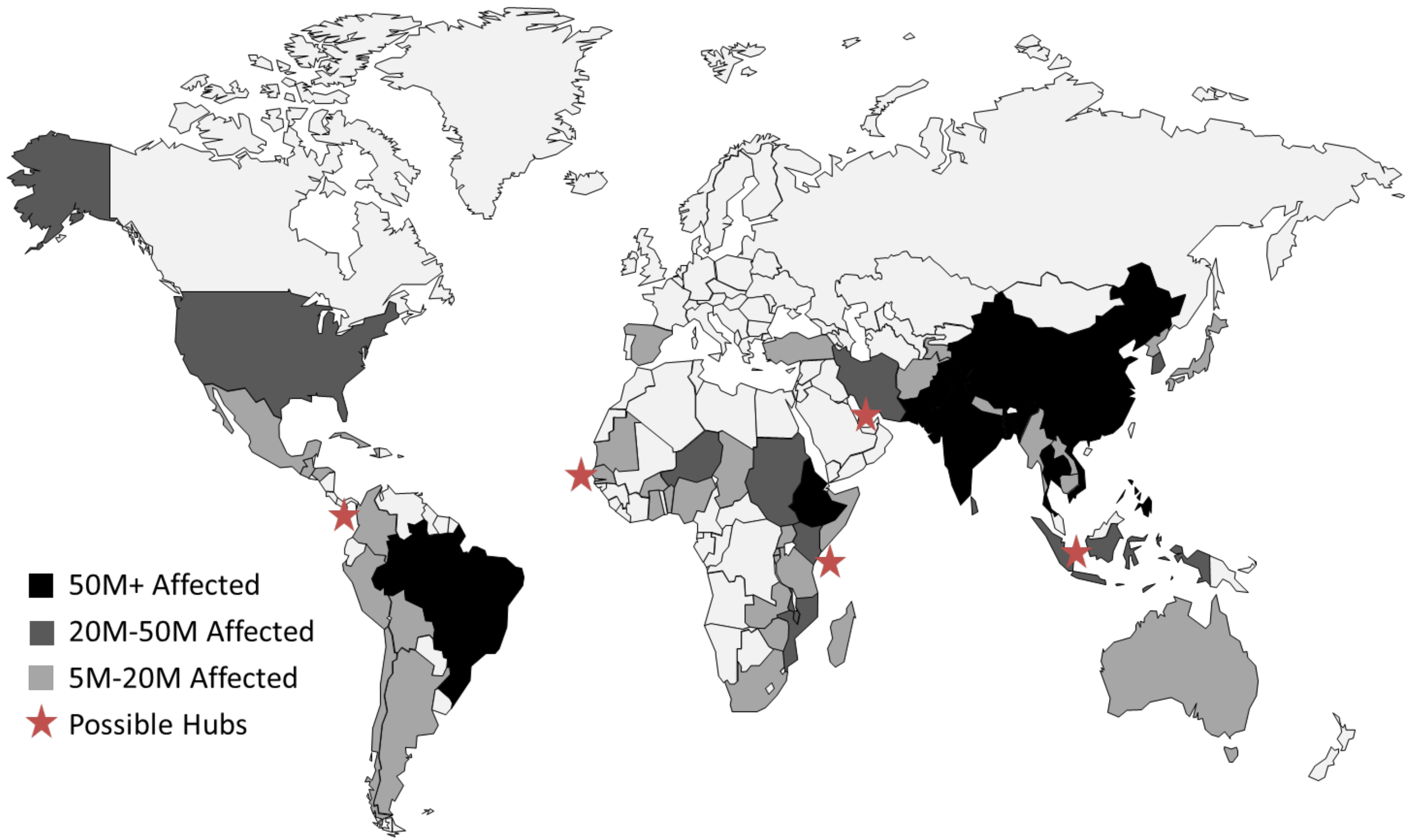

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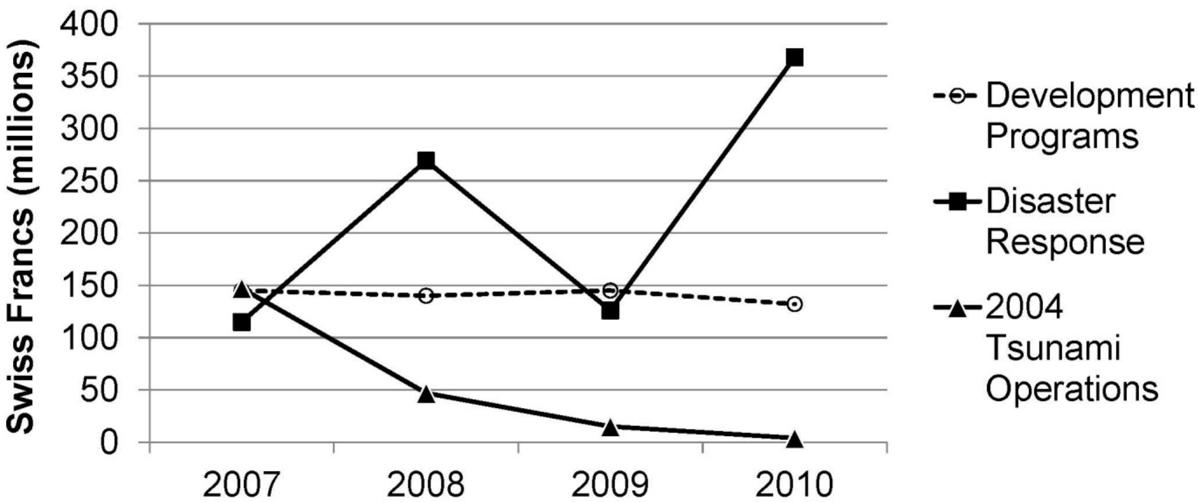
Exhibit 2

IHO Donations



Note: A majority of the 2010 Disaster Response donations were earmarked for Haiti Earthquake relief.

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Exhibit 3a

Number of Four-wheel Drive (4WD) Rental Assignment Vehicles from 2008-2010

(Rental assignment vehicles are the vehicles rented out to National Societies at each point in time)

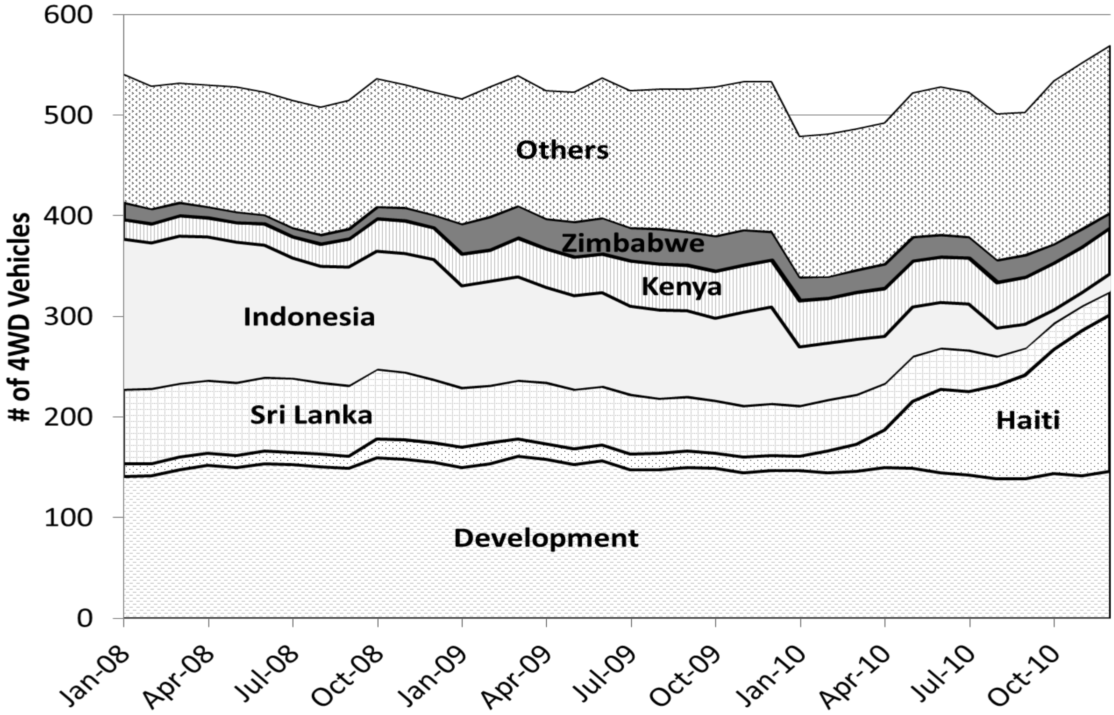

Exhibit 3b

Number of Four-wheel Drive (4WD) Administrative Vehicles from 2008-2010

(Administrative vehicles are the vehicles in-transit to/from a hub, under major repair or outfitting, held for resale, or held in reserve for future disasters)

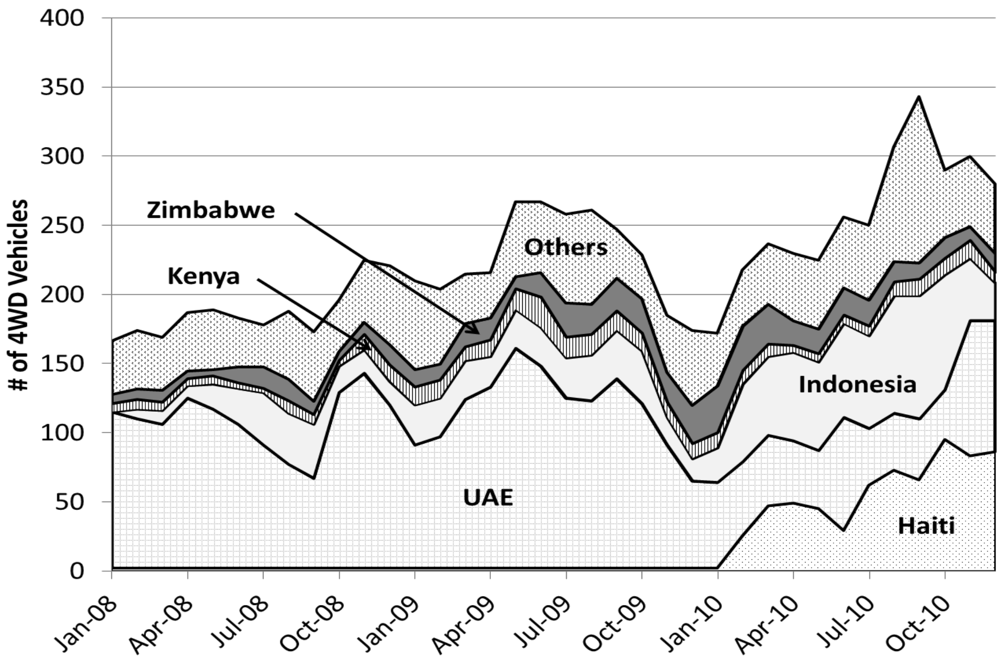

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Exhibit 4

Four-wheel Drive (4WD) Rental Vehicle Ramp-up/down after Mega Disasters

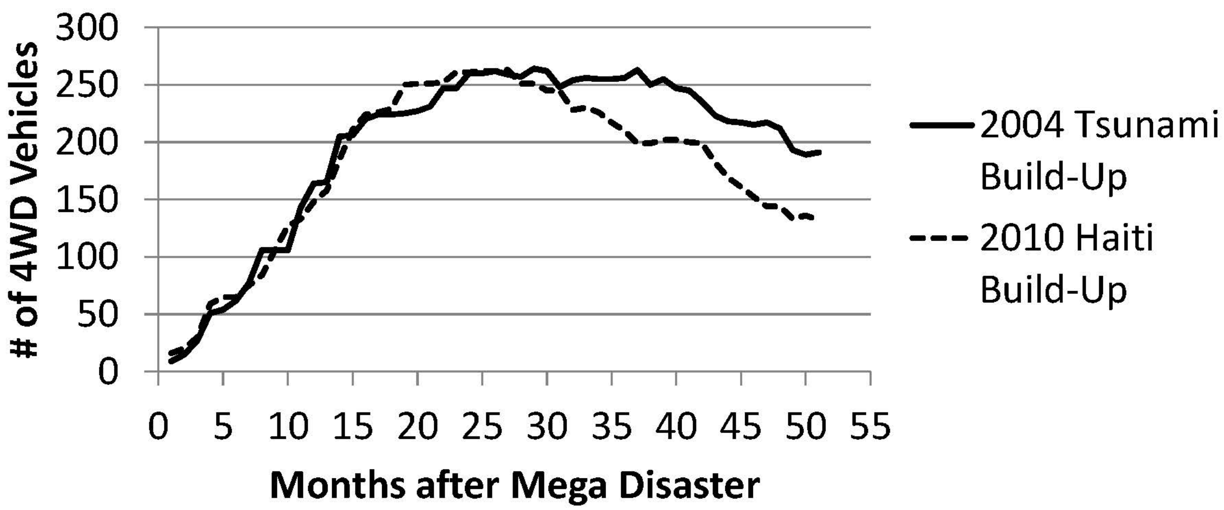

Exhibit 5

Fixed Hub Locations Used in Alex's Simulation Analysis

1-Hub	2-Hub	3-Hub	4-Hub	5-Hub
UAE	UAE	UAE	UAE	UAE
	Panama	Panama	Panama	Panama
		Indonesia	Indonesia	Indonesia
			Senegal	Senegal
				Kenya

Exhibit 6

Optimized Data Results for 4 Supply Chain Configurations

Period	Optimized Transport Costs (excluding Haiti after 2010 earthquake) — Hub Network Configuration				Optimized Haiti Transport Costs (after 2010 earthquake) — Hub Network Configuration				Structural Costs (Hub structural cost = $10,000/month) — Hub Network Configuration				Expediting Costs - 10 Vehicles (Flying in a vehicle= $10,000/month) — Hub Network Configuration			
	5-Hub	3-Hub	1-Hub	1-Hub+TempH	5-Hub	3-Hub	1-Hub	1-Hub+TempH	5-Hub	3-Hub	1-Hub	1-Hub+TempH	5-Hub	3-Hub	1-Hub	1-Hub+TempH
1	72,521	67,643	101,793	101,793	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
2	108,775	113,231	109,607	109,607	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
3	55,813	62,978	58,141	58,141	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
4	62,189	74,228	76,249	76,249	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
5	11,139	5,708	5,708	5,708	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
6	10,250	7,052	7,052	7,052	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
7	50,201	33,471	32,872	32,872	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
8	51,587	52,883	54,823	54,823	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
9	49,477	54,663	68,585	68,585	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
10	65,870	74,244	83,856	83,856	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
11	36,149	34,418	34,418	34,418	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
12	82,466	83,686	83,686	83,686	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
13	45,375	51,176	61,640	61,640	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
14	26,090	21,293	21,293	21,293	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
15	29,476	33,786	44,246	44,246	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
16	129,060	125,110	141,950	141,950	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
17	19,935	18,254	21,294	21,294	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
18	40,524	51,347	57,821	57,821	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
19	28,636	35,729	36,602	36,602	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
20	15,789	17,603	28,272	28,272	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
21	53,149	59,319	41,935	41,935	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
22	28,324	30,621	30,621	30,621	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
23	49,279	53,023	54,062	54,062	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
24	20,624	21,589	25,776	25,776	0	0	0	0	50,000	30,000	10,000	10,000	0	0	0	0
25	45,816	42,219	47,854	47,593	91,760	91,760	91,760	92,021	50,000	30,000	10,000	10,000	4,010	4,010	100,000	100,000
26	2,660	6,412	6,412	6,412	116,175	116,175	198,200	115,500	50,000	30,000	10,000	20,000	0	0	0	0
27	57,385	65,307	75,079	74,729	55,764	55,764	95,136	55,440	50,000	30,000	10,000	20,000	0	0	0	0
28	8,435	19,414	24,310	24,310	116,175	116,175	198,200	115,500	50,000	30,000	10,000	20,000	0	0	0	0
29	42,095	36,046	38,074	37,813	130,116	130,116	221,984	129,360	50,000	30,000	10,000	20,000	0	0	0	0
30	4,460	5,915	5,009	5,009	101,998	101,998	161,056	101,512	50,000	30,000	10,000	20,000	0	0	0	0
31	16,786	18,815	19,081	14,137	70,106	70,106	123,508	76,548	50,000	30,000	10,000	20,000	0	0	0	0
32	82,858	86,477	96,429	96,168	204,468	204,468	348,832	203,280	50,000	30,000	10,000	20,000	0	0	0	0
33	15,633	26,682	27,721	26,579	69,705	69,705	118,920	69,300	50,000	30,000	10,000	20,000	0	0	0	0
34	45,300	42,504	88,284	84,482	92,881	92,881	155,220	95,028	50,000	30,000	10,000	20,000	0	0	0	0
35	110,310	112,124	114,615	114,615	0	0	0	0	50,000	30,000	10,000	20,000	0	0	0	0
36	30,206	32,324	33,622	29,926	0	0	0	2,066	50,000	30,000	10,000	20,000	0	0	0	0

Exhibit 7

Simulation Results for Supply Chain Savings when Changing the Number of Mega Disasters

Table shows the percentage of savings over the baseline 1-Hub configuration.

# of Mega Disasters	1-Hub	2-Hub	3-Hub	4-Hub	5-Hub	1-Hub+ TempH	2-Hub+ TempH	3-Hub+ TempH	4-Hub+ TempH	5-Hub+ TempH
2	Baseline	4.44%	4.01%	0.99%	-3.56%	6.89%	8.33%	4.35%	1.32%	-3.23%
3	Baseline	4.56%	6.46%	3.73%	-0.37%	9.15%	10.02%	6.92%	4.17%	0.07%
4	Baseline	4.86%	8.24%	5.76%	2.08%	10.81%	11.33%	8.75%	6.26%	2.58%
10	Baseline	5.80%	15.61%	14.04%	11.75%	17.88%	17.69%	16.67%	15.08%	12.80%
16	Baseline	6.25%	18.76%	17.60%	15.98%	20.98%	20.75%	20.08%	18.89%	17.25%

Note: negative savings percentages indicate extra costs

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Exhibit 8

Simulation Results for Supply Chain Savings with Two Mega Disasters
when Changing the Average Minor Disaster & Development Program Monthly Demand

Table shows the percentage of savings over the baseline 1-Hub configuration.

Avg. Monthly Demand	1-Hub	2-Hub	3-Hub	4-Hub	5-Hub	1-Hub+ TempH	2-Hub+ TempH	3-Hub+ TempH	4-Hub+ TempH	5-Hub+ TempH
16.4	Baseline	0.02%	-1.80%	-9.45%	-18.73%	6.53%	4.07%	-1.40%	-9.05%	-18.33%
20.3	Baseline	4.15%	1.79%	-4.49%	-11.69%	6.00%	7.55%	2.12%	-4.16%	-11.36%
23	Baseline	4.44%	4.01%	0.99%	-3.56%	6.89%	8.33%	4.35%	1.32%	-3.23%
25.6	Baseline	6.92%	5.69%	2.32%	-1.88%	6.52%	10.25%	5.99%	2.61%	-1.59%
29.4	Baseline	7.21%	7.71%	7.57%	6.05%	7.29%	11.03%	8.12%	7.86%	6.32%
31.9	Baseline	8.99%	8.75%	8.17%	6.72%	6.65%	12.30%	9.06%	8.40%	6.94%
36	Baseline	9.23%	10.26%	12.16%	12.73%	7.77%	13.15%	11.09%	12.52%	12.95%
38.4	Baseline	10.61%	10.94%	12.31%	12.81%	7.25%	14.02%	11.64%	12.63%	13.01%
42.5	Baseline	10.51%	12.00%	15.35%	17.43%	8.27%	14.76%	13.41%	15.96%	17.65%
45	Baseline	11.96%	12.83%	15.76%	17.74%	8.20%	15.78%	14.08%	16.24%	17.92%

Note: negative savings percentages indicate extra costs